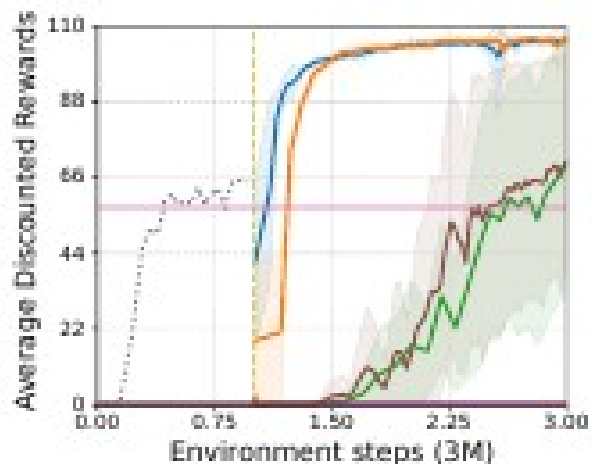


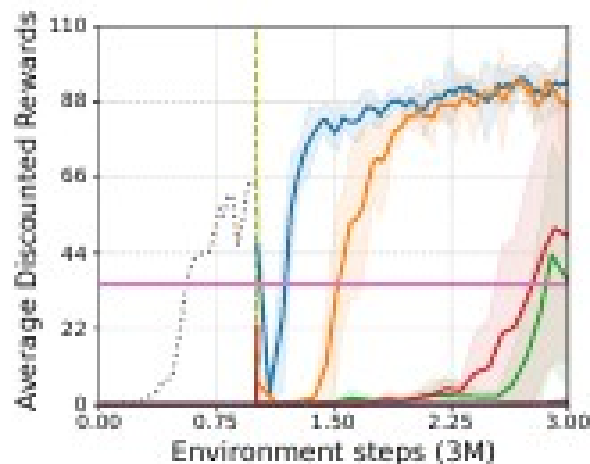
Crisp mechanism.

Not foundation-model prehistory.

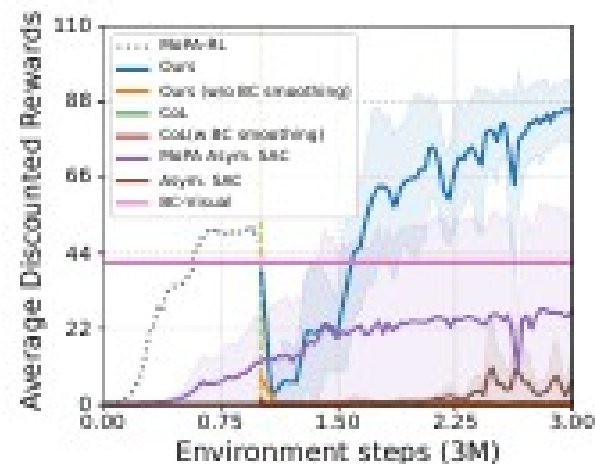
Grade MoPA-PD as an Alternative-approaches classic: privileged teacher, causal ablations, sim-to-sim DR. Do not upgrade the claim.



MoPA-PD



MoPA-PD



MoPA-PD

Five facts that do not change with the seat

Train privilege

01

Classical MP (RRT-Connect) lives inside MoPA-RL (Yamada et al., CoRL 2020). Geometry and the planner are train-time infrastructure, not the shipped controller.

Distill, then smooth

02

Visual BC on direct-action rollouts removes the planner and smooths RRT jitter. MoPA-PD then runs asymmetric SAC: actor from BC, critic from MoPA-RL Q , $Re:R\pi = 1:3$, small α .

Deploy is planner-free pixels

03

Inference is a 32×32 image plus joints. No RRT-Connect, no object state, no motion planner. The critic's state privilege ends when training ends.

Numbers you must be able to say

04

Push 100 / 32.0 / 110.8 (ASR / AEL / ADR). Assembly 0 without BC-smooth vs 100 with. $DR \geq 96.7$ sim-to-sim (Lift Scenario 1 is worst). Expert Push AEL 111 vs Ours 32.

What this paper is not

05

Not an ancestor of ACT, Diffusion Policy, π_0 , OpenVLA, or OXE — contrast poles only. Not a JEPA / LeWM / Cosmos child. Sim-only Sawyer. Do not launder DR into sim-to-real.

Five claims. Only some are equally hard.

1

Deploy without the planner

Distilling MP-augmented state RL via BC and guided vision RL removes the planner and full state at inference.

2

Smoothing is load-bearing

Expert buffer R_e holds BC trajectories, not raw MoPA-RL jitter. Assembly is the test.

3

The efficiency stack

Weight init, expert buffer at 1:3, and a small entropy α — versus Asym. SAC, MoPA-Asym. SAC, and CoL.

4

DR transfers in sim

Texture, color, and lighting in training. Zero-shot distractors and appearance. Above 96% ASR.

5

Better paths than the teacher

Lower AEL and higher ADR than state MoPA-RL. Not merely a clone of the expert.

Vision-from-scratch and “MP inside vision” both zero out

	Sawyer Push			Sawyer Lift			Sawyer Assembly		
	ASR ↑	AEL ↓	ADR ↑	ASR ↑	AEL ↓	ADR ↑	ASR ↑	AEL ↓	ADR ↑
MoPA-RL	98.2	111.0	52.7	95.0	109.2	52.7	99.8	63.3	82.4
BC-Visual	99.4	118.0	46.9	62.0	108.8	34.6	97.0	115.1	50.4
Asym. SAC	0.0	250.0	0.0	0.0	250.0	0.2	0.0	250.0	3.4
MoPA-Asym. SAC	0.0	250.0	0.0	0.0	250.0	0.0	0.0	250.0	0.0
CoL	0.0	250.0	0.0	29.8	173.3	9.7	0.0	250.0	0.0
CoL w/ BC smoothing	0.0	250.0	0.0	0.0	250.0	5.1	0.0	250.0	0.0
Ours w/o BC smoothing	100.0	34.0	108.7	99.4	43.6	100.7	0.0	250.0	0.0
Ours	100.0	32.0	110.8	99.0	42.0	101.7	100.0	61.7	84.5

Table 1: Average success rate (ASR), episode length (AEL), and discounted return (ADR) of our method and baselines averaged over five seeds. Each method is evaluated after 3M environment steps. MoPA-RL [1] is the expert agent trained with state-based policy. All baselines below horizontal line are trained with asymmetric actor-critic for fair comparison. Maximum horizon is 250 for all tasks.

Read rows, not only the bold last line. MoPA-Asym. SAC keeps the planner and still scores 0. BC-Visual keeps ASR on Push and Assembly and loses the path. Without smoothing, Assembly is 0.

Sample efficiency is visible, and it is not just ASR

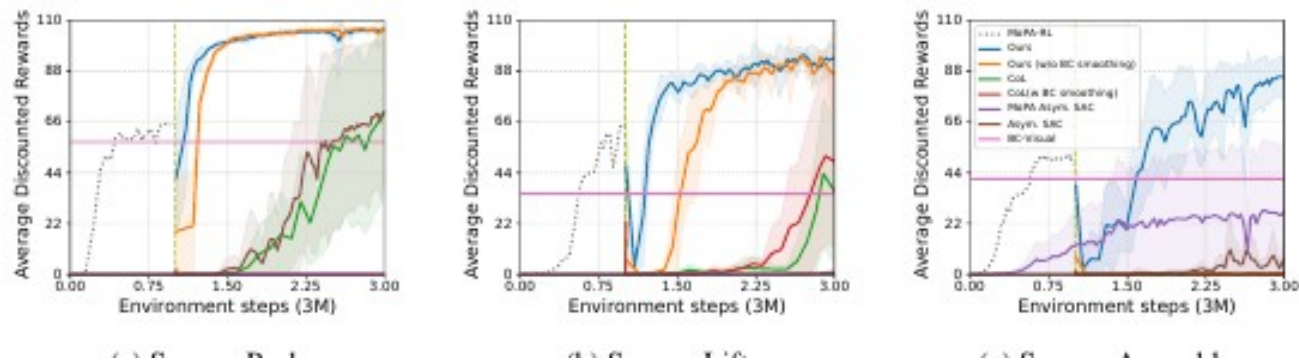


Fig. 3

Discounted return, 0–3M. Ours on top. The handoff off the state teacher is the vertical mark.

Path quality

Push AEL 32.0 vs expert 111.0 and vs BC 118.0. ADR 110.8 vs 52.7 vs 46.9.

Lift needs Stage 2

BC-Visual ASR 62.0. Ours 99.0. Cloning is not enough when coverage is thin.

Related work

Neural motion planners that stay state-based are distinguished, not quietly absorbed. That honesty holds.

Assembly is 0 without it and 100 with it

WITHOUT BC SMOOTH

0

Assembly ASR

WITH BC SMOOTH

100

Assembly ASR

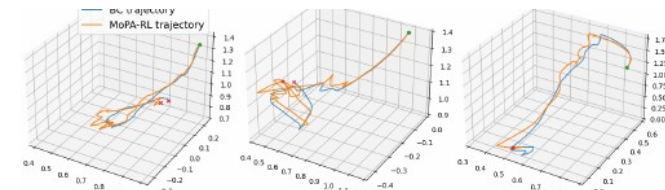


Fig. 8 — same start (green) and goal (red). Orange MoPA-RL paths jitter. Blue BC paths do not. Push and Lift still learn from raw replay, slower. Assembly does not.

Push still works

100 / 34.0 / 108.7 without smoothing vs 100 / 32.0 / 110.8 with. The knife is not Push.

Why jitter hurts

RRT prioritizes collision avoidance over smoothness. Raw paths in Re make RL difficult and, on Assembly, impossible.

Without init, success stays near zero past a million steps

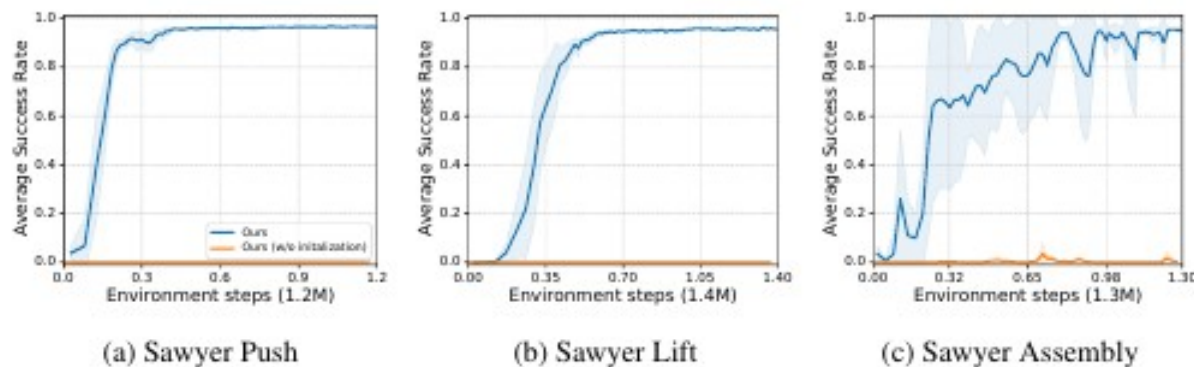


Figure 7: Learning curves comparing our method with and without actor-critic initialization for all three environments. Without initialization, the agent achieves minimal or no success rate, whereas it quickly learns to achieve optimal performance with weight initialization across all the tasks.

With init

Actor $\leftarrow$ BC. Critic $\leftarrow$ MoPA-RL Q. Success $\approx$ 1.0 well before 1.0M steps, all three tasks.

Without init

“Minimal or no success” up till 1.2M steps. No successes observed until then.

Why init is legal

MoPA-RL actions superset the direct action space, so the critic transfer is defined.

Still privilege

That critic was trained with state. Init carries the privilege into the visual run.

Not mere cloning. Also not a free lunch.

	Environment Steps					
	0.2M			1.0M		
	ASR ↑	AEL ↓	ADR ↑	ASR ↑	AEL ↓	ADR ↑
MoPA-RL [1]	69.2	102.7	41.0	98.0	105.4	54.5
BC-Visual	93.2	137.4	37.1	98.8	57.4	97.2
Ours	100.0	34.4	108.2	100	32.0	110.8

(a) Sawyer Push

	Environment Steps								
	0.5M			0.52M			1.0M		
	ASR ↑	AEL ↓	ADR ↑	ASR ↑	AEL ↓	ADR ↑	ASR ↑	AEL ↓	ADR ↑
MoPA-RL [1]	41.4	153.8	21.0	60.6	138.3	29.2	95	109.2	52.7
BC-Visual	48	189.4	14.1	55.2	146.6	24.2	63	109.4	34.8
Ours	99.0	42.0	101.6	99.4	42.9	101.0	99.0	42.0	101.7

(b) Sawyer Lift

Table 4: Average success rate (ASR), episode length (AEL), and discounted return (ADR) of our method compared with MoPA-RL and BC using sub-optimal and optimal MoPA-RL models. Even with sub-optimal training, when our state-based agent does not learn to completely solve the task, our method for distillation into a visual policy achieves high performance and successfully solves the

Push @ 0.2M

Expert ASR 69.2. Ours 100.0, AEL 34.4. BC-Visual is 93.2 / 137.4.

Lift @ 0.5M

Expert ASR 41.4. Ours 99.0, AEL 42.0. BC stays at 48.

Read carefully

The visual phase is still 2M steps. A weak teacher is not “no teacher.” Init still uses that critic.

Authors

Does not heavily depend on the state agent’s success. It does benefit from the initialization.

External validity stops at the simulator door

No real robot

Transfer claims are sim-to-sim. §5 names a real fine-tune and a vision critic as future work. Writing “sim-to-real” rewrites the paper.

One embodiment, three tasks

Sawyer only. Push, Lift, Assembly. The third uses the IKEA furniture env. Household generalization is not measured.

32×32 RGB

Enough to prove the recipe. Not evidence about wrist cameras or multi-view stacks. That external validity is unknown.

2021 baseline scope

CoL and asymmetric SAC are the foils. They zero out. That does not adjudicate ACT, Diffusion Policy, or any VLA.

α is a knob. CoL is not saved by smoothing.

Entropy α

Teacher final log α (Table 3): Push -0.63 , Lift -2.7 , Assembly -0.29 . Visual stage uses smaller α . Sweep $\{0.5, 0, -3, -10, -20, -40\}$. Smaller is stabler. No per- α ASR table.

CoL is the LfD foil

Best CoL cell is Lift ASR 29.8. BC smoothing does not rescue it (Lift ASR 0, ADR 5.1). Do not over-read a 2020 baseline as a 2025 one.

BC model selection

Fig. 9: low test MSE is not the best policy. They take the earliest epoch with 100% validation success. Min-MSE is a different method.

MoPA-Asym. SAC = 0

Keeping RRT in the visual action loop fails all three tasks. Distillation is not optional sugar. Collapsing the plan into a reactive policy is the result.

Accept the recipe. Refuse the upgrade.

ALTERNATIVE APPROACHES CLASSIC

The privileged-teacher claim holds in obstructed simulation. The ablations that matter are in the paper. The generalizations people will want are not.

Accept

Planner removal. BC-smooth knife. Init effect. Path-length gain. Sim-to-sim DR above 96%.

Do not accept

Real clutter solved. Ancestor of π_0 or OpenVLA. Child of JEPA or Cosmos. A BEHAVIOR baseline.

Four probes that do not require new experiments

Where is the planner?

At inference, architecturally absent. Ask for a log of planner calls anyway. The PDF does not print that column.

Whose success?

Push ASR flatters BC. Make the authors (and yourself) answer in AEL and ADR, then look at Lift and Assembly.

Whose baseline?

CoL and Asym. SAC are 2020-era foils. Ask what would change if the BC baseline were a chunked or generative policy. The paper cannot answer. That is fine. Pretending it did is not.

Whose reality?

Table 2 changes textures and adds distractor cubes. Ask which sentence in §5 they want remembered. It is the future-work sentence.

The sentence to leave in the notes

Table 1 decides the mechanism. Assembly 0 vs 100 decides smoothing. Fig. 7 decides init. Table 4 says a weak expert can still teach. Nothing in the PDF decides reality.

Push 100 / 32.0 / 110.8. Expert AEL 111 vs 32. DR worst cell 96.7, sim-to-sim. Not an ancestor. Not a world model.